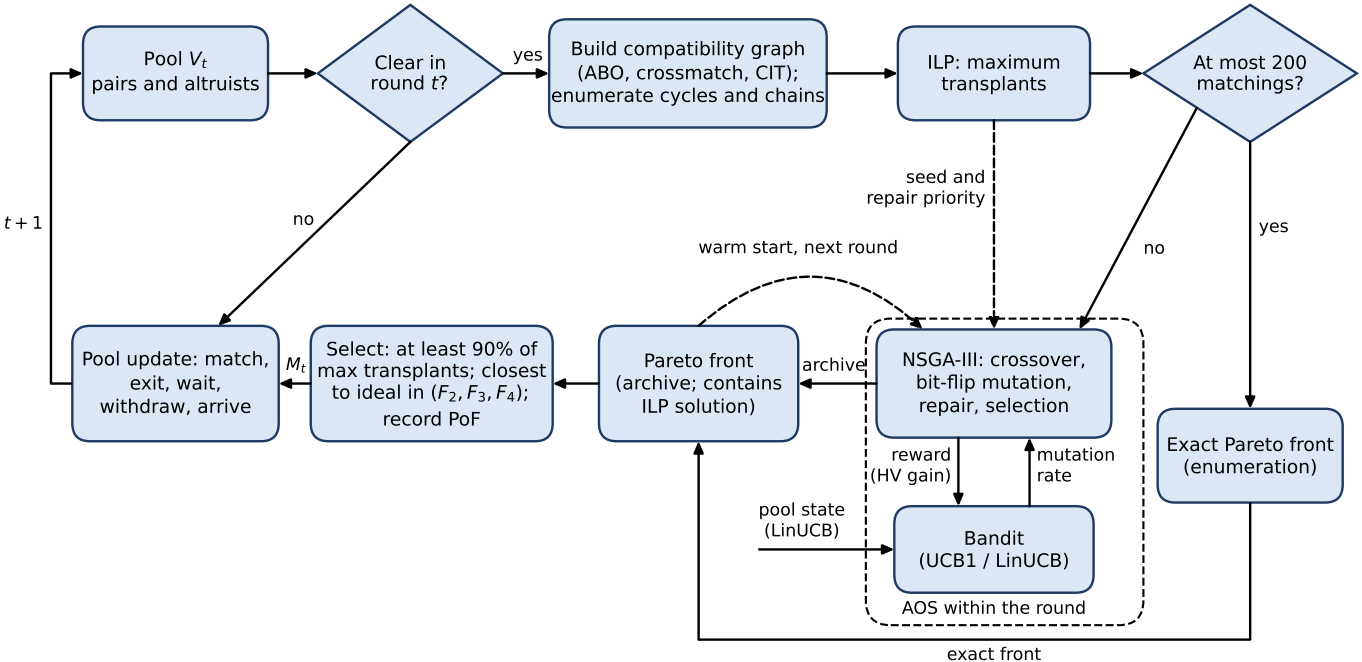
Pool $V_t$ pairs and altruists
Clear in round $t$?
yes
Build compatibility graph (ABO, crossmatch, CIT); enumerate cycles and chains
ILP: maximum transplants
At most 200 matchings?
$t+1$
no
seed and repair priority
warm start, next round
no
yes
Pool update: match, exit, wait, withdraw, arrive
$M_t$
Select: at least 90% of max transplants; closest to ideal in $(F_2, F_3, F_4)$; record PoF
Pareto front (archive; contains ILP solution)
archive
NSGA-III: crossover, bit-flip mutation, repair, selection
Exact Pareto front (enumeration)
reward (HV gain)
mutation rate
pool state (LinUCB)
Bandit (UCB1 / LinUCB)
AOS within the round
exact front